

CariSurg MedTech Pathways Research Project

Preliminary Proposal

Project Title: AI-Assisted Emergency Triage — Towards Prospective Validation and Clinician Trust

Week 1 Final Submission

Statement of the Problem

Emergency department triage remains inconsistent and resource-intensive, placing patients at risk through both over-triage and under-triage. The evidence is clear: machine learning models outperform human nurses in predictive accuracy. Yet no study has prospectively validated an ML triage model in a live clinical setting, and no structured mechanism exists to build the clinician trust that sustained adoption requires. Algorithmic bias and poor data quality compound the problem. This pilot will test a deployable, trustworthy AI triage system that bridges the gap between research performance and real-world clinical use.

Previous Work

The following summaries draw on seven peer-reviewed papers published between 2024 and 2025. Each covers the clinical or engineering problem, the method used, the reported outcome, and the authors' own stated limitation.

Tyler et al. (2024)

Emergency departments in the United States process approximately 131 million visits annually, yet conventional triage relies on the subjective judgement of the triage nurse, which varies across institutions, shift patterns, and individual experience. Tyler et al. conducted a scoping review of 29 peer-reviewed studies published between 2018 and 2023, identified through a systemised search of EMBASE, Ovid MEDLINE, and Web of Science, to examine whether machine learning could improve triage accuracy and consistency. Across all 29 studies, machine learning models outperformed conventional triage systems in predictive accuracy, disease identification, and risk assessment; one AI tool, KATE™, correctly assigned triage acuity in 75.7% of cases compared to 59.8% for triage nurses. The authors acknowledge that every study included in the review was

retrospective and drawn from a single institution, which substantially limits how far the findings can be applied to other hospital settings or real-time clinical environments.

Da'Costa et al. (2025)

Overcrowding, resource constraints, and inconsistency in patient prioritisation place emergency departments under persistent pressure, and traditional triage systems — which depend heavily on subjective clinical judgement — deteriorate further during peak hours or mass casualty events. Da'Costa et al. conducted a narrative review of peer-reviewed articles published between 2015 and 2024, retrieved from PubMed, Scopus, IEEE Xplore, and Google Scholar, to examine the key components, benefits, and barriers of AI-driven triage systems. The review found that AI improves patient prioritisation and reduces wait times; one emergency department reported a 30% reduction in average wait time after integrating a real-time AI system that adjusted prioritisation automatically based on patient vitals and department capacity. However, the authors identify data quality issues, algorithmic bias, lack of clinician trust, and unresolved ethical concerns as significant barriers that continue to block widespread adoption.

Araouchi & Adda (2024)

An ageing population and growing case complexity place increasing strain on triage, which must prioritise patients by severity under conditions of limited resources and rising demand. Araouchi and Adda developed and evaluated TriageIntelli, a system built on seven machine learning models — including Support Vector Machines, Random Forests, Artificial Neural Networks, Gradient Boosting Machines, Logistic Regression, XGBoost, and a stacking ensemble — trained to predict patient triage levels using the Korean Triage and Acuity Scale. The stacking model, which combined the outputs of all seven algorithms, produced the strongest predictive performance, demonstrating that ensemble approaches consistently outperform any single model in triage classification. The authors note that the system was validated on a single dataset using one triage scale only, and that further testing across different health systems and clinical frameworks is required before the approach can be considered deployable at scale.

Tahernejad et al. (2024)

Conventional triage systems were designed for routine hospital settings and struggle to scale during mass casualty events such as natural disasters, where patient volumes spike suddenly and resources are critically constrained. Tahernejad et al. conducted a PRISMA-compliant systematic review, searching PubMed, Scopus, and Web of Science without a time limit through September 2024, to investigate how AI and technology can support triage during emergencies and disasters and to identify the challenges of implementing intelligent triage systems in those conditions. The review found that AI-assisted tools including smart glasses integrated with TensorFlow, mobile e-triage systems, and deep learning models improved both the speed and accuracy of casualty assessment during mass casualty incidents compared to conventional methods. The authors note that most included studies were conducted in high-income countries under controlled conditions, limiting the generalisability of findings to resource-limited or low-income settings where disasters disproportionately occur.

Yi, Baik & Baek (2025)

Despite a growing body of retrospective evidence, the real-world clinical performance of AI-based triage systems remains poorly understood, as the majority of studies have not been tested in live, prospective settings. Yi, Baik, and Baek conducted a PROSPERO-registered systematic review of prospective studies, searching CINAHL, Cochrane, Embase, PubMed, and ProQuest, to identify and evaluate the effects of AI-based triage applied in real-time clinical conditions. Across seven studies meeting inclusion criteria, AI-assisted triage consistently reduced overtriage and undertriage, with model accuracy ranging from 80.5% to 99.1%, and a feed-forward neural network achieving 33% precision in Level 1 triage classification. The authors conclude that while AI shows clear potential as a supportive tool for triage nurses, the limited number of prospective studies and significant variation in methodology means that broader clinical validation across different settings and populations remains an urgent priority.

Chang et al. (2024)

Triage decisions in busy emergency departments are frequently subjective, vary between clinicians, and rely exclusively on structured data such as vital signs and triage scores, ignoring the rich clinical information contained in free-text nursing notes. Chang et al. developed machine

learning models that combined structured triage data with natural language processing of bilingual free-text notes, training and internally validating the models on data from China Medical University Hospital and externally validating them at Asia University Hospital. All ML models achieved higher F1 scores than emergency physicians making disposition decisions based on the same triage data alone, with the Random Forest model selected for its robust calibration and predictive performance. The authors acknowledge that the retrospective design and use of data from two Taiwanese hospitals limits generalisability to other clinical contexts, and that the models were evaluated only for disposition outcome rather than full triage level classification.

Hinson et al. (2025)

Conventional ED triage has well-documented associations with inequitable outcomes, with biased decision-making consistently observed across race, ethnicity, and spoken language in multiple health systems. Hinson et al. conducted a secondary analysis of data collected during a multisite quality improvement initiative at Johns Hopkins and Yale, measuring the effect of outcomes-driven AI-informed decision support on the equity of triage decision-making across sites. The study found that AI-informed triage support reduced both under-triage and over-triage and improved the consistency and equity of triage decisions across racial and ethnic groups. The authors note the study's observational design and relatively short implementation period as limitations, and call for longer-term prospective evaluation to confirm whether equity gains are sustained at scale.

Gap Analysis

Across seven papers, AI-assisted triage consistently outperforms conventional methods in controlled settings. Two specific, actionable gaps remain unresolved in the literature and are directly addressable within a 12-week pilot.

Gap 1 — Absence of prospective, multi-site validation

Tyler et al. (2024) found that all 29 studies in their review were retrospective and conducted at single institutions. Yi, Baik, and Baek (2025) confirm this, identifying only seven prospective studies globally and noting substantial methodological variation between them. Chang et al. (2024)

and Araouchi and Adda (2024) both acknowledge that their models were validated on single datasets and cannot yet be considered generalisable. Hinson et al. (2025) represent a rare exception — a multisite implementation — but their study focused on equity outcomes rather than prospective predictive performance. No study has tested whether ML triage models hold their performance advantage in a live, prospective, single-site setting outside the existing evidence base. A 12-week pilot at Mercer General could address this directly by running an ML model in parallel with existing triage, logging real-time predictions against clinical outcomes, and producing the first prospective performance data from this institution.

Gap 2 — No validated mechanism for building clinician trust in AI triage

Da’Costa et al. (2025) identify lack of clinician trust as one of the primary barriers to AI triage adoption, yet none of the studies reviewed by Tyler et al. (2024) tested any structured approach to building or measuring that trust. Tahernejad et al. (2024) reinforce this in the disaster context, noting that staff familiarity and confidence in the system determined uptake regardless of technical performance. Hinson et al. (2025) demonstrate that equity improvements are achievable through AI-supported triage, but do not report a mechanism for sustaining clinician engagement beyond the quality improvement period. The gap is operational: an AI system that clinicians distrust or override systematically cannot improve outcomes regardless of its accuracy. A 12-week pilot implementing a structured disagreement-logging protocol — in which clinicians record instances where they override the AI recommendation and briefly explain why — would produce both a clinician trust baseline and actionable data for model refinement that no existing study has yet generated in a live ED setting.

Proposed Solution

I propose a 12-week pilot study at Mercer General Hospital that deploys an ML-based triage model in parallel with the existing triage workflow in the emergency department. The system will ingest the same patient data currently collected at triage — including vital signs, age, presenting complaint, and clinical observations — and generate a real-time risk classification alongside the nurse-assigned triage level. The AI recommendation will be visible to the triage nurse but will not override clinical judgement. Every instance in which the nurse disagrees with the AI classification

will be logged with a brief rationale, generating a structured dataset of disagreement patterns over the pilot period.

This design addresses both gaps identified in the literature. Running the model prospectively in a live clinical setting fills the validation gap identified by Tyler et al. (2024), Yi, Baik, and Baek (2025), and multiple individual studies. The disagreement-logging protocol addresses the trust barrier identified by Da’Costa et al. (2025) and Tahernejad et al. (2024), producing measurable data on clinician confidence rather than treating trust as an unquantifiable obstacle. Incorporating free-text nursing notes — as demonstrated by Chang et al. (2024) — will enhance model performance beyond what structured vital signs alone can achieve, while the equity monitoring framework drawn from Hinson et al. (2025) will ensure the system does not replicate existing biases in triage decision-making. By the end of 12 weeks, the pilot will have produced a prospective performance baseline, a clinician trust profile, and a prioritised set of model refinement targets.

References

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